

Exercise 3:

Assuming tables:

```
1  CREATE TABLE Customers (  
2      CustID NUMBER PRIMARY KEY,  
3      Name VARCHAR2(100),  
4      Age NUMBER,  
5      Balance NUMBER(10,2),  
6      IsVIP VARCHAR2(5)  
7  );  
8  
9  INSERT INTO Customers VALUES (101, 'John', 65, 15000, 'FALSE');  
10 INSERT INTO Customers VALUES (102, 'Alice', 45, 8500, 'FALSE');  
11 INSERT INTO Customers VALUES (103, 'Bob', 72, 22000, 'FALSE');  
12 INSERT INTO Customers VALUES (104, 'Emma', 30, 12000, 'FALSE');  
13 INSERT INTO Customers VALUES (105, 'David', 61, 9500, 'FALSE');  
14  
15 COMMIT;
```

```
17 CREATE TABLE Accounts (  
18     AccountID NUMBER PRIMARY KEY,  
19     CustID NUMBER,  
20     AccountType VARCHAR2(20),  
21     Balance NUMBER(12,2),  
22     FOREIGN KEY (CustID) REFERENCES Customers(CustID)  
23 );  
24  
25 INSERT INTO Accounts VALUES (1001, 101, 'Savings', 25000);  
26 INSERT INTO Accounts VALUES (1002, 102, 'Savings', 18000);  
27 INSERT INTO Accounts VALUES (1003, 103, 'Current', 30000);  
28 INSERT INTO Accounts VALUES (1004, 104, 'Savings', 12000);  
29 INSERT INTO Accounts VALUES (1005, 105, 'Current', 15000);  
30  
31 COMMIT;
```

```

33 CREATE TABLE Employees (
34     EmployeeID NUMBER PRIMARY KEY,
35     EmployeeName VARCHAR2(50),
36     Department VARCHAR2(30),
37     Salary NUMBER(10,2)
38 );
39
40 INSERT INTO Employees VALUES (1, 'Raj', 'IT', 50000);
41 INSERT INTO Employees VALUES (2, 'Priya', 'HR', 45000);
42 INSERT INTO Employees VALUES (3, 'Kiran', 'IT', 60000);
43 INSERT INTO Employees VALUES (4, 'Anjali', 'Finance', 55000);
44 INSERT INTO Employees VALUES (5, 'Rohit', 'HR', 40000);
45
46 COMMIT;

```

Scenario 1:

```

48 CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
49 AS
50 BEGIN
51     UPDATE Accounts
52     SET Balance = Balance + (Balance * 0.01)
53     WHERE AccountType = 'Savings';
54
55     COMMIT;
56
57     DBMS_OUTPUT.PUT_LINE('Monthly interest processed');
58 END;
59 /
60
61 BEGIN
62     ProcessMonthlyInterest;
63 END;
64 /

```

Output:

Monthly interest processed

Scenario 2:

```
66 CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(  
67     dept IN VARCHAR2,  
68     bonusPercent IN NUMBER  
69 )  
70 AS  
71 BEGIN  
72     UPDATE Employees  
73     SET Salary = Salary + (Salary * bonusPercent / 100)  
74     WHERE Department = dept;  
75  
76     COMMIT;  
77  
78     DBMS_OUTPUT.PUT_LINE('Employee bonus updated');  
79 END;  
80 /  
81  
82 BEGIN  
83     UpdateEmployeeBonus('IT', '10');  
84 END;  
85 /
```

Output:

Employee bonus updated

Scenario 3:

```
87 CREATE OR REPLACE PROCEDURE TransferFunds(  
88     fromAcc IN NUMBER,  
89     toAcc IN NUMBER,  
90     amount IN NUMBER  
91 )  
92 AS  
93     bal NUMBER;  
94 BEGIN  
95     SELECT Balance INTO bal  
96     FROM Accounts  
97     WHERE AccountID = fromAcc;  
98  
99     IF bal >= amount THEN  
100         UPDATE Accounts  
101         SET Balance = Balance - amount  
102         WHERE AccountID = fromAcc;  
103  
104         update Accounts  
105         SET Balance = Balance + amount  
106         WHERE AccountID = toAcc;  
107  
108         COMMIT;  
109  
110         DBMS_OUTPUT.PUT_LINE('Transfer Successful');  
111  
112     ELSE  
113         DBMS_OUTPUT.PUT_LINE('Insufficient Funds');  
114     END IF;  
115 END;  
116 /  
117  
118 BEGIN  
119     TransferFunds(1001,1002,5000);  
120 END;  
121 /
```

Output:

Transfer Successful